

1. Introduction

In this paper, I describe the methods used for data analysis and present brief results following the structure of a research article. This exploratory data analysis is part of a broader study. The original study examined whether the control (no reflection) and intervention groups (reflection prompted after each lecture) showed significant differences in learning achievement. For the intervention group, I also explored the relationships between learning achievement and reflection-related variables, including self-reported levels of confusion and interest and the specificity of reflections on confusing and interesting parts.

In addition to such analyses, this exploratory data analysis aims to investigate differences in reflection content among students with different grade levels. Specifically, I compared three groups: A, B, and C. Such groups were decided based on the students' final grades, calculated as a weighted score of exams and assignments. My focus is on identifying how the content of their reflections differed. To analyze the text data, I used the Linguistic Inquiry and Word Count (LIWC) tool, a widely used text analysis program that categorizes words based on psychological meaning (Tausczik & Pennebaker, 2010). Given the exploratory nature of this study, I opted to use the pre-existing, validated dictionaries provided by LIWC rather than developing a custom dictionary.

The LIWC analysis has not been actively used in STEM education. However, I found this method useful in analyzing reflection data that can provide insight into whether students' learning has occurred as intended. Especially in the context of undergraduate STEM education, many science and engineering courses are taught in lecture-based format, especially in introductory courses where hundreds of students enroll. Still, just with traditional instruction tools, promoting reflection-on-action in a large classroom can be challenging due to the logistics involved in collecting a large number of student reflections and assessing the relevance and quality of each reflection in a timely manner. By using digital technology accessible to individual students and instructors, such as a mobile application, individual reflection activities can be incorporated into a course more easily.

While my previous studies focused on the specificity of student reflections, I found it useful to apply computational analysis like LIWC to the student reflections, seeing the potential of such a method allowing a deeper understanding of students' reflection writing patterns. More specifically, it can reveal how high achievers and low achievers differ in their language use, which in turn helps educators better understand their students' thought processes and provides opportunities for more customized support.

2. LIWC Dictionaries and Hypothesis

In this section, I briefly summarize four major dictionaries that will be referenced in the results section. The descriptions are cited from LIWC website¹.

- ‘Analytical Thinking (Analytic)’ reflects how formal, logical, and structured a person’s thinking is based on their language use. High scores in this dictionary are often rewarded in academic settings and are linked to better reasoning skills. In contrast, low scores indicate a more intuitive and personal language style, which can feel warmer and more personable.
- ‘Clout’ reflects the social confidence or leadership displayed in someone's language. It measures the speaker’s status or authority in social interactions, but it differs from ‘Power,’ which focuses on awareness of social hierarchy.
- ‘Authenticity’ measures how genuine or spontaneous a person's language is. High Authenticity scores are linked to unfiltered, honest communication, often found in casual or intimate settings. Lower scores are typical in prepared or socially cautious language.
- ‘Emotional Tone (Tone)’ summarizes the overall emotional tone of a text, combining both positive and negative dimensions. Scores above 50 indicate a positive tone, while those below 50 suggest negativity.

After learning the LIWC analysis process and dictionaries used in such a process, I hypothesized that students in the A Group would score highest across most LIWC dictionaries, followed by B Groups, while those in the C Group would score the lowest. This expectation was based on prior findings that the C Group consisted of a smaller population of students who were less engaged and whose reflections tended to be more vague. Such findings suggested that the C Group would show lower scores in categories, especially in analytic thinking and clarity of expression, reflecting their lower abilities in writing logically and clearly about what confuses and/or interests them.

3. Method

a. Participant

This study involved 94 engineering and science students enrolled in an introductory physics course at a public university in the Northeastern part of the United States. The Institutional Review Board (IRB) approved the recruitment process. Only students who

¹ <https://www.liwc.app/help/liwc#Summary-Measures>

consented to share their grade information were included in the data analysis. Throughout the academic semester, students were asked to reflect on their learning after each lecture, but it was not a mandatory assignment in the course.

b. Data Collection

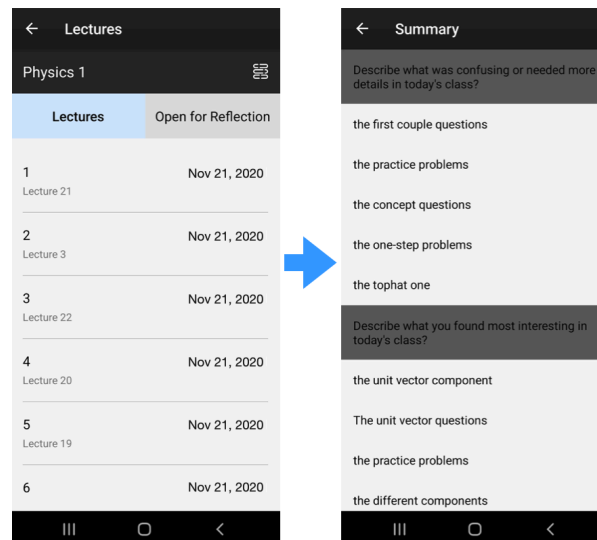


Figure 1. An example screenshot of the CourseMIRROR app.

Reflection data was gathered using the mobile application 'CourseMIRROR' (Mobile In-situ Reflections and Review with Optimized Rubrics). This application was developed to prompt and collect students' self-reflections and feedback during any course. It captures student reflections to help instructors identify areas students struggle and find intriguing and provide additional support throughout the semester.

Figure 1 illustrates the app's user interface. On the left, students could select a lecture from their enrolled course. After selecting a lecture, they were prompted to answer two reflection questions: "What was confusing or needed more detail in today's class?" and "What did you find most interesting in today's class?" Students were given a specific timeframe to submit their reflections. Both students and instructors could view a summarized list of responses for these questions, as shown on the right side of Figure 1. Data was collected continuously throughout the semester.

Table 1. Number of reflections on confusing and interesting parts by student groups

	Number of reflections on confusing parts	Number of reflections on interesting parts	Word counts of the entire reflections on confusing parts	Word counts of the entire reflections on interesting parts
A Group (A+, A, A-)	384	384	6829	6155
B Group (B+, B, B-)	395	395	8017	6863
C Group (C+, C, C-, F)	65	65	1338	982

c. Data Analysis

The first step in the data analysis was preprocessing the dataset. The raw data contained one reflection record per student, with each row including the student ID, final grade, lecture number, reflection on confusing parts, reflection on interesting parts, and other variables. To compare the three grade groups (A, B, and C), I created a new dataset with six rows. The first three rows combined the confusing reflections from the A, B, and C Groups, respectively. The next three rows combined the interesting reflections from the three student groups. The R code for this step is included in Appendix 1.

Next, I conducted a LIWC analysis using the application provided at LIWC.app². I used the preprocessed dataset and selected the LIWC-22 dictionary, the most recent version developed in 2022. Given the exploratory nature of this analysis, I opted to include all the available dictionaries in this group. After running the analysis, I downloaded the results. The results included score data only without text. As described on the LIWC website³, most LIWC output variables represent the percentage of total words in the text. For instance, a Positive Emotions (emo_pos) score of 4.20 indicates that 4.20% of the words in the text are positive emotion words.

In the final step, I visualized the score data. I created two dot plots: one for reflections on confusing parts and one for reflections on interesting parts. The x-axis represented the types of dictionaries described briefly in Section 2. The y-axis showed the scores for each group. Before creating the visualizations, I deemed dictionaries with scores below 10 % insignificant to analyze, so I excluded them. Because the reflections were specifically focused on confusion and interest, the range of word diversity was already

² <https://www.liwc.app/>

³ <https://www.liwc.app/help/liwc>

narrower compared to other text datasets, such as journals or longer written paragraphs. Then I calculated the variances in scores among the three groups and ordered the dots based on these variances to emphasize the dictionaries that showed the most and least differences among the grade groups. The R code for this step is included in Appendix 2.

4. Result

The LIWC analysis generated a data frame containing the dictionary scores. For visualization, I considered both bar graphs and dot plots at first and also tried to include the numerical scores in the graphs for accuracy. However, since 15 and 17 dictionaries were used in the confusing and interesting reflection analyses, bar graphs could not effectively display all the information. Therefore, I opted for dot plots, as they allowed the numerical data to be presented more clearly despite the slightly cluttered appearance.

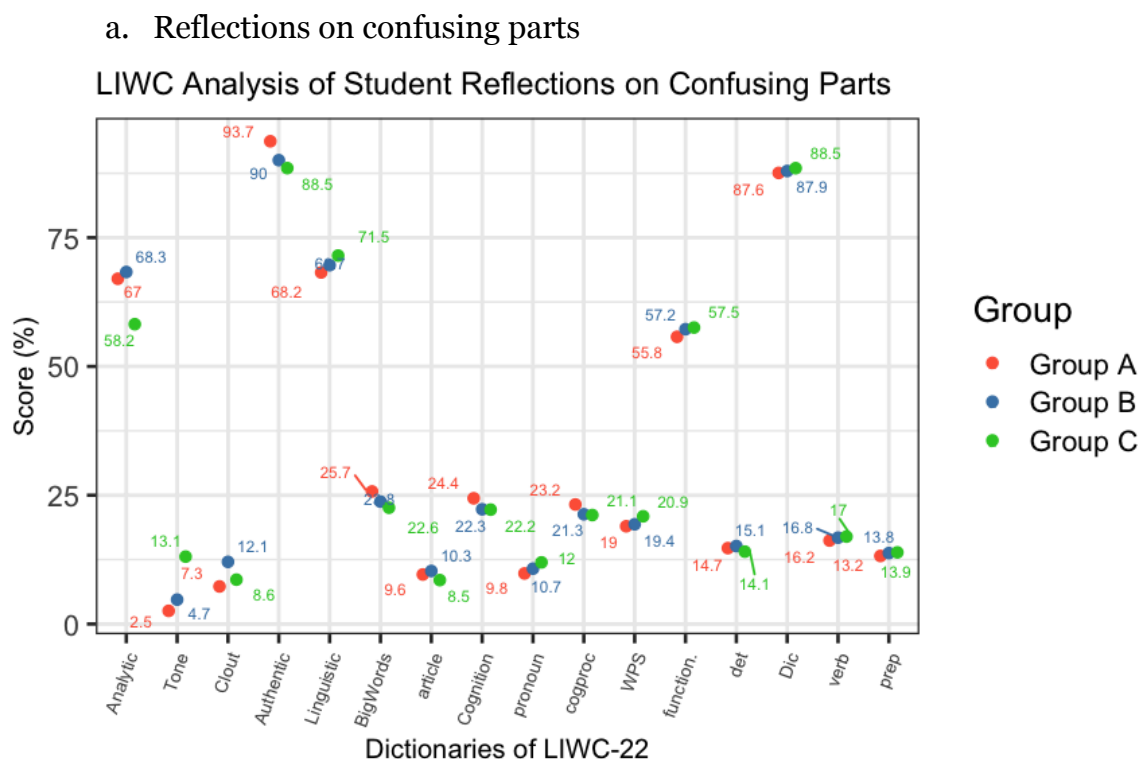


Figure 2. The result of LIWC analysis of reflections on confusing parts

Figure 2 shows how the reflections on confusing parts differ among the three student groups based on LIWC dictionaries. Significant differences were observed in five

dictionaries: 'Analytic,' 'Tone,' 'Clout,' 'Authentic,' and 'Linguistic.' These dictionaries showed the largest between-group variations.

- Analytic: A Group scored 67.0, slightly lower than B Group (68.3) but higher than C Group (58.2). Despite the lack of differences between the A and B Groups, I found that the C Group, a relatively low-achieving group of students, tends to use analytic words less frequently than their mid and high-achieving students.
- Tone: The C Group exhibited notably more positive emotions compared to the A and B Groups. While not shown in the graph, the tone is generally neutral to negative. C Group had the most positive tone (13.1), whereas B Group's tone (4.7) and A Group's tone (2.5) were much lower. This was in contrast with my broad hypothesis.
- Clout: All groups scored low on Clout, with B Group scoring the highest (12.1) and A Group the lowest (7.3). This is because all the reflections were about confusing parts of the lecture, which is often a negative aspect of their learning experiences.
- Authenticity: All groups scored high in authenticity, with A Group at 93.7, B Group at 90.0, and C Group at 88.5. This implies that most participants' responses were worth to be analyzed.
- Linguistic Metrics: Although 'Linguistic Metrics' were included in the visualization, I could not deem between-group differences significant.
 - WPS (Words Per Sentence): C Group had the highest average of 20.9 words per sentence, followed by B Group (19.4) and A Group (19.0), suggesting that C Group tended to write longer sentences when reflecting on confusing parts.
 - BigWords (Percentage of words longer than six letters): A Group had the highest percentage of complex vocabulary (25.7), followed by B Group (23.8) and C Group (22.6).
 - Dictionary Words (Dic): All groups had similar scores, with A Group at 87.6, B Group at 87.9, and C Group at 88.5, indicating almost 90 % of their words were recognized by the LIWC dictionary.

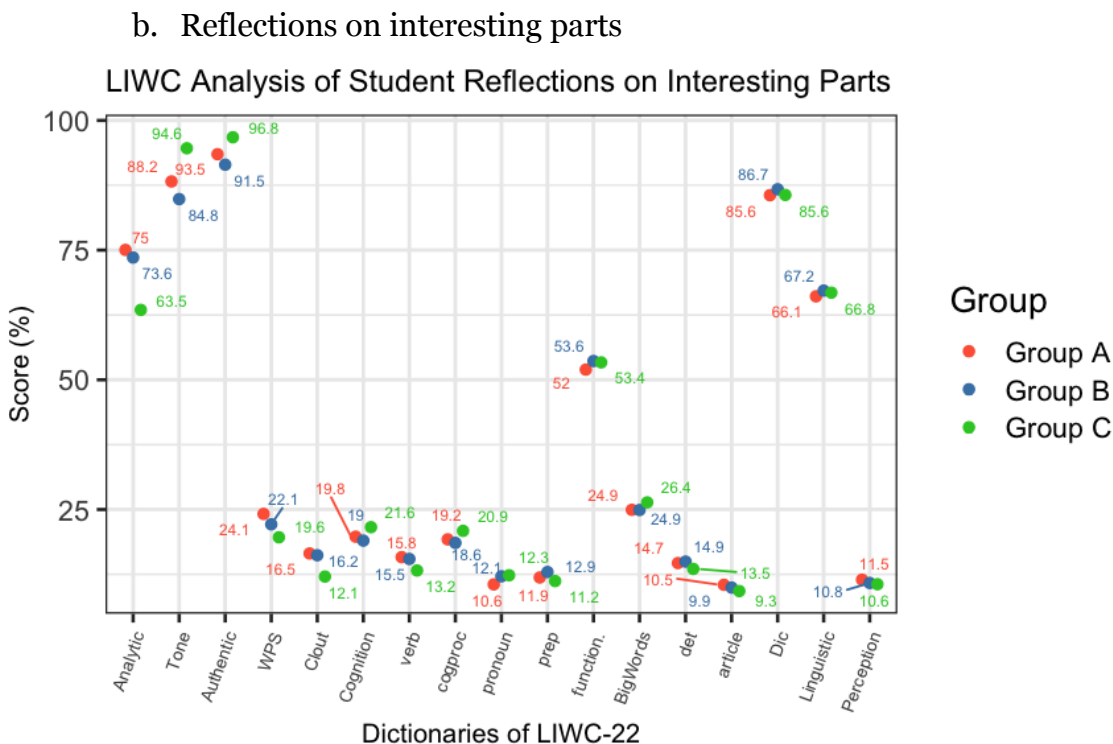


Figure 3. The result of LIWC analysis of reflections on interesting parts

Figure 3 shows how the three student groups' reflections on interesting parts differ based on LIWC dictionaries. Significant differences were found in the 'Analytic,' 'Tone,' 'Authentic,' 'WPS,' and 'Clout' dictionaries, while other dictionaries showed no notable between-group variations.

- Analytic: The A Group had the highest analytic score at 75.0, followed by the B Group at 73.6 and the C Group at 63.5. High-achieving students tended to use analytics words more frequently than low-achieving students. Such a pattern was similar to that observed in reflections on confusing parts. Regardless of the objects of reflection, I observed the positive relationships between the frequency of analytic word usage and learning achievement despite the absence of statistical significance.
- Tone: The tone was notably more positive in reflections on interesting parts than confusing parts, which is expected due to the nature of the reflection prompts (interest is generally associated with positive emotions, whereas confusion reflects a more negative state of learning). Unlike the hypothesis, C Group exhibited the most positive tone at 94.6, with A Group at 88.2 and B Group at 84.8.

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- **Authenticity:** Authenticity scores remained consistently high across groups, similar to the patterns seen in reflections on confusing parts. Interestingly, Group C showed the highest score of authenticity despite the small between-group differences.
 - **WPS (Words Per Sentence):** The pattern here was opposite to that seen with reflections on confusing concepts, where C Group had the highest WPS. In this case, A Group had the highest WPS at 24.1, followed by B Group at 22.1 and C Group at 19.6.
 - **Clout:** Clout scores were much higher for reflections on interesting parts than confusing parts, likely because Clout is related to confidence, and confidence is more closely associated with interest than confusion. A Group (16.5) and B Group (16.2) showed similar Clout levels. C Group showed the lowest level (12.1).

5. Conclusion and Self-Reflection

Given the limited dataset and the specificity of the reflections, which focused on confusing and interesting aspects of physics concepts, the observed between-group differences were not particularly striking. However, consistent with my broad hypothesis, I found a positive relationship between the frequency of analytic word usage and learning achievement. Such a finding suggests that students with higher academic performance tend to engage in more structured, logical thinking. While the differences in this exploratory analysis were subtle, the relationship between word usage patterns and learning outcomes is an intriguing area for further investigation. A more comprehensive study with a larger dataset could offer deeper insights into these patterns, particularly through the use of statistical methods like regression analysis. By employing such methods, it would be possible to quantify the extent to which the use of analytic language predicts academic success.

Another interesting finding was that the C Group expressed more emotions in their reflections than the other two groups. This trend was consistent in both confusing and interesting reflections. While all three groups scored low on emotional expression in reflections about confusing parts, they scored higher when reflecting on interesting parts. I inferred that low-achieving students might underestimate their level of confusion, suggesting they might be less objective about their understanding than high-achieving students. Additionally, the level of interest students feel in class does not necessarily relate to their academic achievement. Such two findings, although not strongly evidenced in the current analysis, lead to the need for educators to support low-achieving students in accurately assessing what confuses them and the extent of their confusion. Furthermore, fostering interest in learning should be encouraged for all

students, regardless of their current academic performance. Even if interest does not immediately translate to improved learning outcomes, it can serve as a catalyst for sustained learning motivation over time.

Although only four to five dictionaries revealed notable between-group differences, LIWC analysis has the potential to be utilized in other STEM education contexts. It could be applied to analyze open-ended metacognitive activities, journal writings, and logical science writing practices. As a STEM education researcher, I found it fascinating to integrate methods commonly used in sociology into the analysis of educational data.

References

Tausczik, Y. R., & Pennebaker, J. W. (2010). The psychological meaning of words: LIWC and computerized text analysis methods. *Journal of Language and Social Psychology*, 29(1), 24–54. <https://doi.org/10.1177/0261927X09351676>

Appendix 1. Data Preparation

Code ▼

Load necessary libraries

Hide

```
library(readxl)
library(dplyr)
```

Data load

Hide

```
# Read the Excel file
file_path <- "Phys174_NVIV0.xlsx"
data <- read_excel(file_path, sheet = "all")
```

Data creation

Hide

```
# Filter the data by grade group
a_group <- data %>% filter(`grade group` == "A group")
b_group <- data %>% filter(`grade group` == "B group")
c_group <- data %>% filter(`grade group` == "C group")

# Combine the reflections for each group
a_group_confusing <- paste(na.omit(a_group$`confusing reflection`),
                           collapse = " ")
b_group_confusing <- paste(na.omit(b_group$`confusing reflection`),
                           collapse = " ")
c_group_confusing <- paste(na.omit(c_group$`confusing reflection`),
                           collapse = " ")

a_group_interesting <- paste(na.omit(a_group$`interesting reflection`),
                             collapse = " ")
b_group_interesting <- paste(na.omit(b_group$`interesting reflection`),
                             collapse = " ")
c_group_interesting <- paste(na.omit(c_group$`interesting reflection`),
                             collapse = " ")

# Create the final dataframe with six rows
df_combined <- data.frame(
  Type = c("A_group_Confusing", "B_group_Confusing", "C_group_Confusing",
           "A_group_Interesting", "B_group_Interesting", "C_group_Interesting"),
  Reflections = c(a_group_confusing, b_group_confusing, c_group_confusing,
                  a_group_interesting, b_group_interesting, c_group_interesting),
  stringsAsFactors = FALSE
)
```

Save as csv file

Hide

```
write.csv(df_combined, file = "reflection_Kim.csv", row.names = FALSE)
```

Check if the file was saved correctly

Hide

```
df_cnew <- read.csv("reflection_Kim.csv",
                   header = TRUE, stringsAsFactors = FALSE)
print(df_cnew)
```

Type

<chr>



A_group_Confusing

B_group_Confusing

Type
<chr>
C_group_Confusing
A_group_Interesting
B_group_Interesting
C_group_Interesting

6 rows | 1-1 of 2 columns

Count the number of non-missing confusing and interesting reflections for each group

Hide

```
a_group_confusing_count <- sum(!is.na(a_group$`confusing reflection`))
b_group_confusing_count <- sum(!is.na(b_group$`confusing reflection`))
c_group_confusing_count <- sum(!is.na(c_group$`confusing reflection`))

a_group_interesting_count <- sum(!is.na(a_group$`interesting reflection`))
b_group_interesting_count <- sum(!is.na(b_group$`interesting reflection`))
c_group_interesting_count <- sum(!is.na(c_group$`interesting reflection`))
```

Create a table summarizing the number of reflections for each group and type

Hide

```
reflection_count_table <- data.frame(
  Group = c("A Group", "B Group", "C Group"),
  Confusing_Reflections = c(a_group_confusing_count,
                             b_group_confusing_count,
                             c_group_confusing_count),
  Interesting_Reflections = c(a_group_interesting_count,
                              b_group_interesting_count,
                              c_group_interesting_count)
)

print(reflection_count_table)
```

Group	Confusing_Reflections	Interesting_Reflections
<chr>	<int>	<int>
A Group	384	384
B Group	395	395

Group	Confusing_Reflections	Interesting_Reflections
<chr>	<int>	<int>
C Group	65	65
3 rows		

Appendix 2. Visualization

[Code ▼](#)

Load necessary libraries

[Hide](#)

```
df_score <- read.csv("LIWC_scoresOnly.csv",  
                    header = TRUE, stringsAsFactors = FALSE)  
  
print(df_score)
```

Type <chr>	Segm... <int>	WC <int>	Analytic <dbl>	Cl... <dbl>	Authentic <dbl>	Tone <dbl>	WPS <dbl>	BigWor... <dbl>	Dic <dbl>	
A_group_Confusing	1	6829	67.02	7.29	93.71	2.55	18.97	25.73	87.57	
B_group_Confusing	1	8017	68.33	12.06	90.04	4.73	19.36	23.76	87.94	
C_group_Confusing	1	1338	58.20	8.61	88.50	13.08	20.91	22.57	88.49	
A_group_Interesting	1	6155	75.04	16.52	93.49	88.24	24.14	24.94	85.59	
B_group_Interesting	1	6863	73.57	16.18	91.47	84.84	22.14	24.90	86.73	
C_group_Interesting	1	982	63.48	12.07	96.76	94.64	19.64	26.37	85.64	

6 rows | 1-10 of 120 columns

Load necessary libraries

[Hide](#)

```
library(dplyr)  
library(tidyr)  
library(ggplot2)  
  
# package to reorder factors  
library(forcats)  
  
# position labels near the points without overlap  
library(ggrepel)
```

Convert df_score to long format, excluding ‘Segment’ and ‘WC’

[Hide](#)

```
df_long <- df_score %>%

# exclude 'Segment' and 'WC(wordcount)' columns
select(-Segment, -WC) %>%

# transform all columns except 'Type' into long format
pivot_longer(cols = -Type,
              names_to = "Dictionary",
              values_to = "Score")

print(head(df_long))
```

Type <chr>	Dictionary <chr>	Score <dbl>
A_group_Confusing	Analytic	67.02
A_group_Confusing	Clout	7.29
A_group_Confusing	Authentic	93.71
A_group_Confusing	Tone	2.55
A_group_Confusing	WPS	18.97
A_group_Confusing	BigWords	25.73
6 rows		

Reflections on confusing parts

Calculate the differences among groups and filter out dictionaries with low scores

Hide

```

confusing_vis <- df_long %>%
  filter(Type %in% c('A_group_Confusing', 'B_group_Confusing',
                    'C_group_Confusing')) %>%
  group_by(Dictionary, Type) %>%
  summarize(mean = mean(Score),
            se = sd(Score) / sqrt(n()),
            N = n(),
            .groups = "drop") %>%

  # reshape a dataset from long format to wide format
  # 'Type' will be columns
  pivot_wider(names_from = Type, values_from = mean) %>%

  # add filtering step to drop dictionaries where all scores are below 10
  filter(`A_group_Confusing` >= 10 | `B_group_Confusing` >= 10 |
        `C_group_Confusing` >= 10) %>%

  # calculate the difference among three groups (variance)
  mutate(diff = abs(`A_group_Confusing` - `B_group_Confusing`) +
            abs(`C_group_Confusing` - `C_group_Confusing`) +
            abs(`B_group_Confusing` - `C_group_Confusing`)) %>%

  # sort by the largest difference first
  arrange(desc(diff))

```

Reorder the Dictionary terms based on the difference

Hide

```

confusing_vis <- confusing_vis %>%

  # make the dataframe long again
  pivot_longer(cols = c(`A_group_Confusing`, `B_group_Confusing`, `C_group_Confusing`),
               names_to = "Type", values_to = "mean") %>%

  # and then reorder based on the between-group variances
  mutate(Dictionary = fct_reorder(Dictionary, diff, .desc = TRUE))

```

Plot the reordered data

Hide


```

# Visualization using ggplot
p_confusing <- confusing_vis %>%

# `fill` sets bar colors by group
ggplot(aes(x = Dictionary, y = mean, color = Type)) +

# use `geom_point` for dot plot, `position = "dodge"` for side-by-side bars
geom_point(position = position_dodge(width = 0.5), size = 2, alpha = 1) +

# color the bars based on the groups
scale_color_manual(name = "Group",
                    values = c("A_group_Confusing" = "#FF6347",
                                "B_group_Confusing" = "#4682B4",
                                "C_group_Confusing" = "#32CD32"),
                    labels = c("Group A", "Group B", "Group C")) +

# add numbers on the top of each bar
geom_text_repel(aes(label = round(mean, 1)),
                position = position_dodge(width = 0.8),
                size = 2.5,
                alpha = 1) +

# black and white theme
theme_bw(base_size = 16) +

# customize axis and label
labs(x = "Dictionaries of LIWC-22", y = "Score (%)",
     title = "LIWC Analysis of Student Reflections on Confusing Parts") +

theme(axis.text.x = element_text(angle = 70, hjust = 1, size = 8),

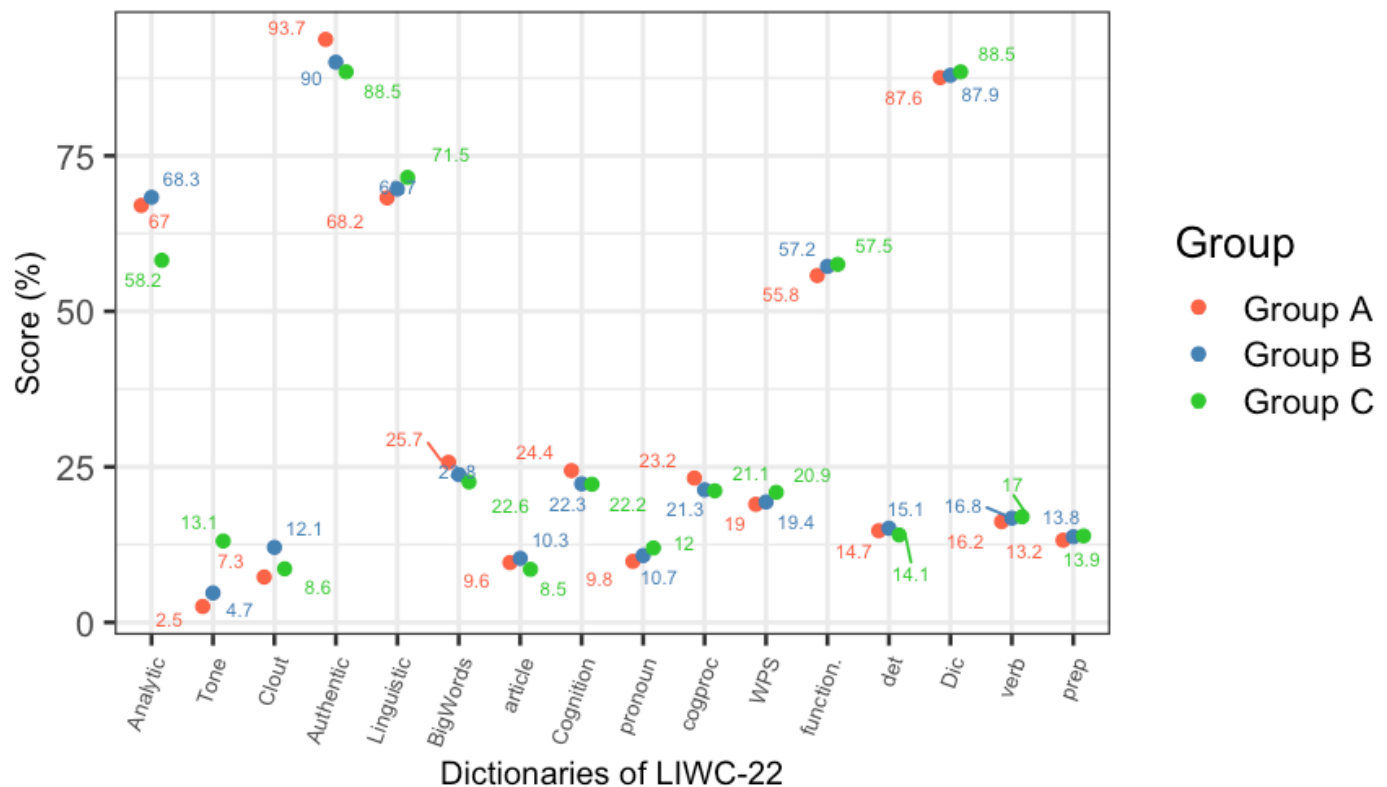
      # adjust the title font size
      plot.title = element_text(size = 14),

      # adjust x,y-axis label font size
      axis.title.x = element_text(size = 12),
      axis.title.y = element_text(size = 12))

p_confusing

```

LIWC Analysis of Student Reflections on Confusing Parts



Reflections on interesting parts

Calculate the differences among groups and filter out dictionaries with low scores

Hide

```
interesting_vis <- df_long %>%
  filter(Type %in% c('A_group_Interesting', 'B_group_Interesting', 'C_group_Interesting')) %>%
  group_by(Dictionary, Type) %>%
  summarize(mean = mean(Score),
             se = sd(Score) / sqrt(n()),
             N = n(),
             .groups = "drop") %>%

  # reshape a dataset from long format to wide format
  # 'Type' will be columns
  pivot_wider(names_from = Type, values_from = mean) %>%

  # add filtering step to drop dictionaries where all scores are below 10
  filter(`A_group_Interesting` >= 10 | `B_group_Interesting` >= 10 |
         `C_group_Interesting` >= 10) %>%

  # calculate the difference among three groups (variance)
  mutate(diff = abs(`A_group_Interesting` - `B_group_Interesting`) +
           abs(`A_group_Interesting` - `C_group_Interesting`) +
           abs(`B_group_Interesting` - `C_group_Interesting`)) %>%

  # sort by the largest difference first
  arrange(desc(diff))
```

Reorder the Dictionary terms based on the difference

Hide

```
interesting_vis <- interesting_vis %>%

  # make the dataframe long again
  pivot_longer(cols = c(`A_group_Interesting`, `B_group_Interesting`,
                        `C_group_Interesting`),
               names_to = "Type", values_to = "mean") %>%

  # and then reorder based on the between-group variances
  mutate(Dictionary = fct_reorder(Dictionary, diff, .desc = TRUE))
```

Plot the reordered data

Hide

```

# Visualization using ggplot
p_interesting <- interesting_vis %>%

# `fill` sets bar colors by group
ggplot(aes(x = Dictionary, y = mean, color = Type)) +

# use `geom_point` for dot plot, `position = "dodge"` for side-by-side bars
geom_point(position = position_dodge(width = 0.5), size = 2, alpha = 1) +

# color the bars based on the groups
scale_color_manual(name = "Group",
                    values = c("A_group_Interesting" = "#FF6347",
                                "B_group_Interesting" = "#4682B4",
                                "C_group_Interesting" = "#32CD32"),
                    labels = c("Group A", "Group B", "Group C")) +

# add numbers on the top of each bar
geom_text_repel(aes(label = round(mean, 1)),
                position = position_dodge(width = 0.8),
                size = 2.5,
                alpha = 1) +

# black and white theme
theme_bw(base_size = 16) +

# customize axis and label
labs(x = "Dictionaries of LIWC-22", y = "Score (%)",
     title = "LIWC Analysis of Student Reflections on Interesting Parts",
     fontsize = 8) +

theme(axis.text.x = element_text(angle = 70, hjust = 1, size = 8),
      plot.title = element_text(size = 14), # Adjust the title font size
      axis.title.x = element_text(size = 12), # Adjust x-axis label font size
      axis.title.y = element_text(size = 12))

p_interesting

```

LIWC Analysis of Student Reflections on Interesting Parts

